

WASP-36b

R-0983

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_181218 · created 2026-07-29T21:41:54+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458470.8417 +/- 0.0032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.092 +/- 0.046
Transit depth (Rp/Rs)^2	0.85 +/- 0.84 [%]
Orbital Inclination (inc)	82.7 +/- 2.94
Airmass coefficient 1 (a1)	1.015 +/- 0.073
Airmass coefficient 2 (a2)	-0.004 +/- 0.017
Scatter in the residuals of the lightcurve fit is	7.13 %
Best Comparison Star	#3 - [216, 131]
Optimal Aperture	13.56
Optimal Annulus	21.46
Transit Duration (day)	0.058 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.092 ± 0.046 vs 0.1368 (deviation 0.89σ)
Duration fit vs published	0.058 d vs 0.0758 d (deviation 0.56σ)
Mid-time O–C	-7.3 min
Depth SNR	1.8
Residual scatter	7.13 %

Light curve

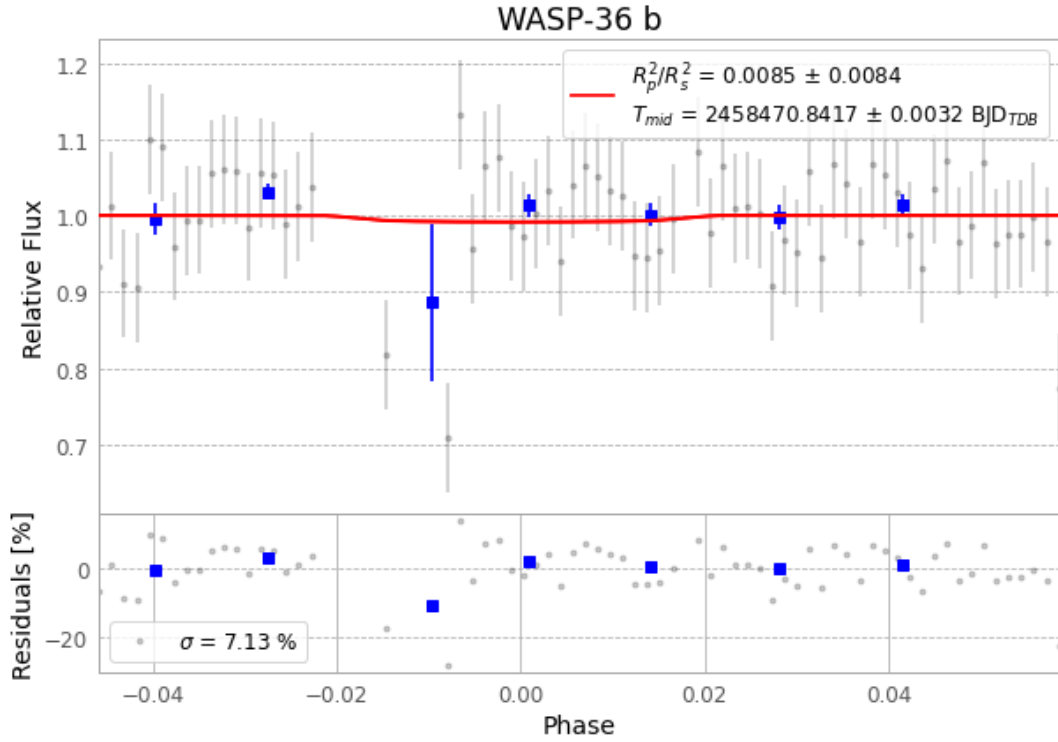


Figure 1 — FinalLightCurve_WASP-36 b_2018-12-17.png (SHA-256 3f06b74abf14a291...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [482, 345], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 21ff8387ee9e7eaf...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

78 FITS frames (51 MB), WASP-36181218062412.FITS → WASP-36181218101512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-181218.log (5fa1c0b4286d...), FinalLightCurve_WASP-36 b_2018-12-17.png (3f06b74abf14...), FinalLightCurve_WASP-36 b_2018-12-17.csv (851843fcd44f...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 21:41Z.